Linear Algebra—Winter/Spring 2026

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Mid-Semester Examination

11:00 AM to 12:30 PM, Saturday, Feb 28, 2026

Instructions

i. *Studied ahead?* Since we have not yet covered some topics in class (e.g. determinants), if you use them, please be sure to define the appropriate objects and prove any property thereof that you use.

ii. *Duration.* You have a total of 90 minutes.

iii. *Do I need to answer all the questions?* A-Type and B-Type questions have options. C-Type questions do not. This is written explicitly inline.

iv. *Points distribution.* A-Type carries 15 points, B-Type carries 16 points. C-Part will contribute to the assignment scaling score.

v. *What if I answer more than the required number of questions?* If, for any given type, you are asked to answer k questions, and you write $k' > k$ answers, then only the first k answers (of that type) will be graded. You are therefore encouraged to only submit k answers (and strike out the rest, should you attempt more than k answers).

A Type—solve any three $(5 \times 3 = 15)$

1. [5 pts] Let V be a vector space over a field F, and let $B = \{\alpha_1, \ldots, \alpha_n\}$ be a fixed set of vectors in V. Suppose that every vector $\alpha \in V$ can be expressed *uniquely* in the form

$$\alpha = \sum_{i=1}^{n} c_i \alpha_i,$$

for some scalars $c_1, \ldots, c_n \in F$. Is B a basis of V (either write down a proof or give a counter-example).

2. [5 pts] Let V be an n-dimensional vector space over a field F, and let $\{\alpha_1, \ldots, \alpha_k\}$ be a set of linearly independent vectors in V, where $k < n$. Let W be a vector space over the same field F, and let $\beta_1, \ldots, \beta_k$ be arbitrary vectors in W. Determine whether there exists a linear transformation $T : V \to W$ such that

$$T(\alpha_j) = \beta_j, \qquad j = 1, \ldots, k.$$

If such a linear transformation exists, is it unique (either write down a proof of uniqueness or give a counter-example)?

3. [5 pts] Let $\mathcal{P}_2$ denote the vector space of all polynomials of degree less than or equal to 2 over the field $\mathbb{R}$. Determine whether the set

$$\{x,\ x+x^2,\ x+x^2+2\}$$

is a basis of $\mathcal{P}_2$.

4. [5 pts] Let V, W be finite-dimensional vector spaces and T be a linear transformation from V to W. Suppose $K = \mathsf{nullspace}(T) \subseteq V$ (i.e. the largest subset of V such that for all $\alpha \in K$, it holds that $T\alpha = \vec{0}$). Fix some $\beta \in \mathsf{range}(T)$. Let $\gamma \in V$ be any vector such that $T\gamma = \beta \neq \vec{0}$. How many linearly independent vectors $\gamma_1, \dots \gamma_n$ exist such that $T\gamma_i = \beta$? Give a proof.

5. Consider the space V of all infinite tuples of real numbers, i.e. let

$$V := \{(x_1, x_2, \dots) : x_1 \in \mathbb{R}, x_2 \in \mathbb{R} \dots\}.$$

Let T be a "left shift operator", i.e. T is a linear transformation that acts as follows: $T(x_1, x_2, \dots) := (x_2, x_3, \dots)$.

   i. [2 pts] Is $\dim(\mathsf{nullspace}(T)) = 0$? Why or why not?

   ii. [2 pts] Is $\mathsf{range}(T) \subsetneq V$? Why or why not?

   iii. [1 pt] In a line, explain how your answers so far are related to the rank-nullity theorem and whether there is any contradiction here, or not.

B Type—solve any two $(8 \times 2 = 16)$

6. Let S be the solution space of the system $RX = 0$, where R is an $m \times n$ row-reduced echelon matrix and X is an $n \times 1$ column vector of variables. Suppose that R has r non-zero rows and that the leading non-zero entries of the non-zero rows occur in columns $k_1, \dots, k_r$ with $k_1 < k_2 < \dots k_r$. Let J be the set consisting of the $n - r$ column indices different from $k_1, \dots, k_r$, i.e., $J = \{1, \dots, n\} \setminus \{k_1, \dots k_r\}$. Then the system $RX = 0$ has the form:

$$x_{k_i} + \sum_{j \in J} c_{ij} x_j = 0, \qquad i \in \{1, 2, \dots, r\},$$

for certain scalars $c_{ij} \in F$.

   i. [3 pts] For each $j \in J$, let E_j be the solution obtained by setting $x_j = 1$ and $x_i = 0$, for all other $i \in J$. Express E_j, $j \in J$ explicitly in terms of the scalars c_{ij}, $1 \leq i \leq r$, $j \in J$.

   ii. [3 pts] Show that S is a vector space and that the vectors E_j, $j \in J$ form a basis of S.

   iii. [2 pts] What is the dimension of the solution space S, i.e. what is $\dim\{X : RX = 0\}$? Prove your assertion.

7. i. [3 pts] Let W_1, W_2 be subspaces such that $W_1 \cap W_2$ is spanned by the basis vectors $\{\alpha_1, \ldots \alpha_k\}$, W_1 is spanned by the basis vectors $\{\alpha_1, \ldots, \alpha_k, \beta_1, \ldots \beta_m\}$ and W_2 is spanned by the basis vectors $\{\alpha_1 \ldots \alpha_k, \gamma_1, \ldots \gamma_n\}$. Show that

$$\{\alpha_1, \ldots \alpha_k, \beta_1, \ldots \beta_m, \gamma_1, \ldots \gamma_n\} \tag{1}$$

form an independent set.

ii. [3 pts] Show that $W_1 + W_2$ is spanned by the vectors (1) above. Using your results so far, find non-zero real numbers c_1, c_2, c_3 and c_4 such that

$$c_1 \cdot \dim(W_1 + W_2) + c_2 \cdot \dim(W_1 \cap W_2) + c_3 \cdot \dim(W_1) + c_4 \cdot \dim(W_2) = 0,$$

for any arbitrary choice of W_1 and W_2. Prove your assertion.

iii. [2 pts] Suppose the vectors $\zeta_1, \ldots \zeta_4$ form a basis for the vector space V, over the field of real numbers (the symbol ζ is the Greek letter 'zeta'). Let

$$V_1 = \mathsf{span}\{\zeta_1, \zeta_3, \zeta_4\}$$

and

$$V_2 = \mathsf{span}\{d_1 \cdot \zeta_1 + d_2 \cdot \zeta_2, d_2 \cdot \zeta_1 - d_1 \cdot \zeta_2, \zeta_3\}$$

where d_1, d_2 are non-zero real numbers. Find (with rigorous justifications), $\dim(V_1 + V_2), \dim(V_1), \dim(V_2), \dim(V_1 \cap V_2)$.

8. Let $T : V \to W$ be a linear transformation from the vector space V to the vector space W. Suppose $\{\alpha_1, \ldots, \alpha_k, \alpha_{k+1}, \ldots \alpha_n\}$ is a basis for V. Figure 1 (see the last page) shows the rank-nullity theorem graphically, for T.

i. [3 pts] Determine what $\mathsf{span}(A)$, $\mathsf{span}(B)$, and $\mathsf{span}(C)$ are in Figure 1. Your options are $\mathrm{Im}(T)$, $\mathsf{nullspace}(T)$ and $\mathrm{Im}(T')$ where $T'(\alpha_i) := \alpha_i$ for $i > k$, and zero otherwise, i.e. $T'(\alpha_1) = T'(\alpha_2) = \cdots = T'(\alpha_k) = 0$.

ii. [3 pts] Suppose $\{\beta_1, \ldots, \beta_\ell\}$ are linearly independent vectors such that

$$\mathsf{span}(\beta_1, \ldots, \beta_\ell) \cap \mathsf{nullspace}(T) = \{\vec{0}\}.$$

Is it the case that $T(\beta_1), \ldots T(\beta_\ell)$ are also linearly independent? If not, give an explicit counter-example. If they are independent, prove it.

iii. [2 pts] Suppose $\mathsf{nullspace}(T) = \{\vec{0}\}$ and $V = W$. Prove or disprove the following: $\mathsf{range}(T) = W$.

C Type—solve both (assignment questions)

9. Suppose V is a vector space that is spanned by a finite set of vectors $\beta_1 \ldots \beta_m$. Show that any independent set of vectors in V must be finite and cannot contain more than m elements.

10. Consider the vectors in $\mathbb{R}^4$ defined by $\alpha_1 = (-1, 0, 1, 2)$, $\alpha_2 = (3, 4, -2, 5)$, $\alpha_3 = (1, 4, 0, 9)$. Find a homogeneous system of linear equations for which the space of solutions is exactly the subspace of $\mathbb{R}^4$ spanned by the three given vectors.

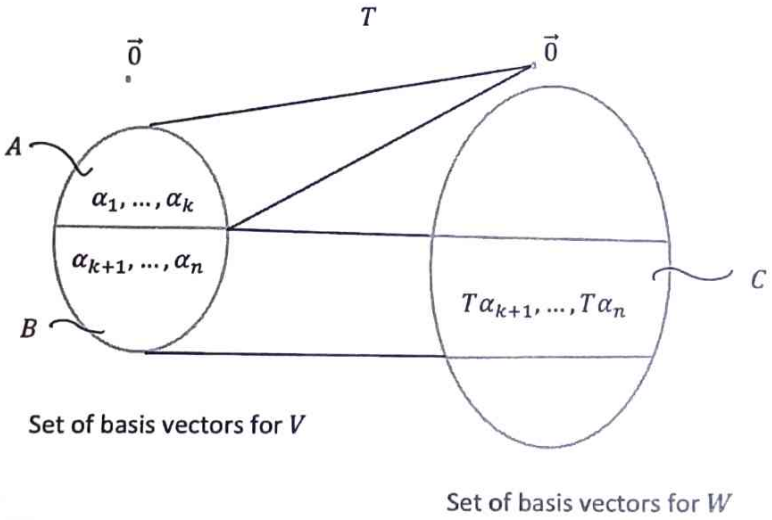

Figure 1: A graphical depiction of the rank-nullity theorem.